Interview Guide: Practical Needs and Challenges in Quantum Software Testing

Semi-structured interviews with quantum computing professionals

Purpose	To understand current testing and validation practices in quantum computing, identify quantum computing professionals' needs and challenges, and assess whether existing quantum software testing (QST) research reflects practical development realities.
Participants	Professionals with experience in quantum computing, including but not limited to quantum simulation, real quantum hardware, or remote access to quantum computers.
Format	Online semi-structured interview via Zoom, with cameras enabled for both the interviewer and interviewee where feasible.
Duration	Approximately 35–50 minutes.

Opening Script and Consent

Thank you for participating in this study. The interview explores your experiences with developing, testing, validating, or operating quantum software and systems. Your responses will be used solely for academic research. With your consent, the session will be recorded to support transcription and analysis. **The recording will not be made public and will only be accessed by the interviewer**, and no personally identifying information will be included in any publication. You may decline to answer any question or stop the interview at any time.

- Do you consent to participate in this interview?
- Do you consent to audio/video recording for research analysis?
- Do you have any questions before we begin?

Part 1. Demographic/Background questions (about 10-15 minutes).

Please discuss the following four aspects in details.

1. **Research focus.**
   - What is your current research or professional focus in quantum computing?

2. **Working environment.**
   - What type of environment do you mainly work in: classical simulation, real

quantum hardware, remote quantum hardware, or a combination of these?
- At what scale do you typically work (e.g., number of qubits, circuit depth, number of gates, or system complexity)?

3. **Years of experience.**
   - How many years of experience do you have with classical simulation, hands-on real quantum hardware, and remote quantum hardware, respectively?

4. **Team size.**
   - What is the typical team size and division of responsibilities for a project in your area?

Part 2. Open-Ended Discussion on QST topics (about 25-35 minutes).

Please choose three topics from the below list and discuss them in details based on your own experience in quantum computing.

Topic 1. Test Oracle Generation

Primary question: How do you obtain the expected output of an execution of a quantum program?

- When are formal or explicit specifications available, and when are they unavailable?
- How do you judge correctness when the expected output is unknown or probabilistic?
- What alternative evidence do you use, such as known properties, repeated executions, classical results, or expert knowledge?
- What should a useful test oracle support?

Topic 2. Debug and Repair

Primary question: How do you localize the cause of the faults of a quantum program and conduct repair?

- Which faults are most common in your work: software defects, hardware defects, configuration errors, calibration problems, or interactions among components?
- How much of the debugging process is manual?
- At what granularity should a tool localize faults (e.g., gate, qubit, circuit component, compiler stage, control system, or hardware subsystem)?
- What information would make fault diagnosis and repair more actionable?

Topic 3. Test Input Generation

Primary question: What constraints exist when selecting inputs for quantum programs?

- What algorithmic constraints determine whether an input state is valid?
- Are some valid states difficult or infeasible to prepare on real hardware?
- Should QST tools check both logical input validity and physical preparability?

Topic 4. Limitations of Real Quantum Computers

Primary question: Which aspects do you think currently limit the practical usage of quantum computers?

- Which hardware imperfections are most important in practice?
- How accurately do existing noise models represent real devices?
- How do observation and measurement limitations, calibration drift, heterogeneous qubit quality, crosstalk, queueing, cost, or limited access influence practical usage of quantum computers?
- What evidence is needed before you consider a result or system reliable?

Topic 5. Simulation-to-Real Gap

Primary question: What discrepancies exist between classical simulation environments and real quantum computers in practice?

- What important behaviors can be studied in simulation but not reproduced on hardware, or vice versa?
- What differences have you observed in feasibility, performance, output distributions, runtime, or resource requirements?
- How should QST tools account for platform- and hardware-specific behavior?

Closing Questions

1. Is there any important practice issue that we have not covered?
2. Would you be willing to be contacted for clarification or follow-up questions?

Closing script: Thank you very much for your time and for sharing your experiences with us. Your insights will be valuable in supporting future research on quantum software testing. Your responses will be anonymized, and no personally identifying information will be included in our publications. Thank you again for your contribution to this study.